

ASHUTOSH DWIVEDI

Python Developer | Flask · FastAPI · Automation · REST APIs

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Python developer who builds practical tools end-to-end — voice-driven AI pipelines, OCR extractors, desktop automation and web backends in clean, testable Python. Strong grasp of OOP, REST API design and packaging. Seeking a Python Developer role to deliver production-quality code.

EDUCATION

B.Tech — Computer Science & Engineering 2023 – 2026 (Expected)
United Institute of Technology, Prayagraj (United University)

TECHNICAL SKILLS

Core Python	OOP, file & exception handling, decorators, virtual environments, packaging
Web & APIs	Flask, FastAPI, REST API design, Jinja2, Streamlit
Key Libraries	psutil, pyttsx3, SpeechRecognition, Pillow, OpenCV, pytesseract, NumPy, pandas
Databases	SQL, SQLite, SQLAlchemy
Tools	Git/GitHub, PyInstaller, VS Code, Postman, Linux

PROJECTS

Voice-to-Prescription — AI Medical Scribe

Tech: Python, FastAPI, Whisper (STT), spaCy NER, Ollama LLM, Streamlit

- Built an end-to-end voice-to-prescription pipeline that converts a doctor's spoken dictation into structured, validated JSON prescriptions — designed to run fully local, with no cloud dependency.
- Integrated Whisper speech-to-text with a spaCy NER + rule-based engine to normalise medicine names, dosages, frequencies and durations; added optional Ollama (Mistral 3B) LLM for higher-accuracy extraction.
- Exposed FastAPI REST endpoints (CORS-enabled) for mobile clients and a Streamlit demo UI with field-level validation and confidence scores.

JARVIS — Personal AI Assistant

Tech: Python, Flask, pywebview, SpeechRecognition, pyttsx3, psutil, OpenAI/Ollama APIs

- Created a JARVIS-style assistant handling voice commands, desktop automation, system monitoring (CPU/RAM via psutil) and conversational replies through OpenAI/Ollama LLM APIs.
- Serves a Flask web interface inside a pywebview desktop shell; packaged a standalone Windows executable using PyInstaller.

ID-Card OCR & Information Extraction

Tech: Python, OpenCV, pytesseract, Pillow, Streamlit, Flask

- Engineered an OCR pipeline that extracts structured fields (name, DOB, ID numbers) from Aadhaar, PAN, driving licence and voter ID images.
- Implemented dedicated per-document extractors with OpenCV/Pillow image pre-processing for higher recognition accuracy; exposed via Streamlit app and Flask API.

AI-Powered Phishing Detection System

Tech: Python, Flask, scikit-learn (Random Forest), JavaScript, HTML, CSS

- Developed a web app that classifies URLs as phishing / legitimate in real time using a Random Forest model trained on lexical URL features with scikit-learn.
- Built the complete stack end-to-end — model training pipeline, Flask REST backend and a responsive cybersecurity-themed frontend (HTML/CSS/JS).

ACHIEVEMENTS & CERTIFICATIONS

- Competed at Rajasthan Police Hackathon 1.0 (RAKAM), Jaipur (Jan 2024)
- Represented Team BLACKBOX 913 at HackQuest '25 — 24-hr hackathon, UIT Prayagraj (Apr 2025)
- Team Cryptoknights at HackDiwas 2024 — 24-hr hackathon, United University (Apr 2024)